Two Measurements per Technician

AP Statistics · Unit 4 · Topic 4.4 · TEACHER KEY

CED TETHER

- **Skill 1.E** — Identify an appropriate inference method for significance tests.
- **Skill 1.F** — Identify null and alternative hypotheses.
- **Skill 4.C** — Verify that inference procedures apply in a given situation.
- **EU VAR-7** — The t-distribution may be used to model variation.
- **LO VAR-7.B** — Identify an appropriate testing method for a population mean with unknown σ, including the mean difference between values in matched pairs. [Skill 1.E]
- **EK VAR-7.B.1** — The appropriate test for a population mean with unknown σ is a one-sample t-test for a population mean.
- **EK VAR-7.B.2** — Matched pairs can be thought of as one sample of pairs. Once differences between pairs of values are found, inference for significance testing proceeds as for a population mean.
- **LO VAR-7.C** — Identify the null and alternative hypotheses for a population mean with unknown σ, including the mean difference between values in matched pairs. [Skill 1.F]
- **EK VAR-7.C.1** — The null hypothesis for a one-sample t-test for a population mean is H_0: $\mu = \mu_0$, where μ_0 is the hypothesized value. Depending upon the situation, the alternative hypothesis is H_a: $\mu < \mu_0$, or H_a: $\mu > \mu_0$, or H_a: $\mu \neq \mu_0$.
- **EK VAR-7.C.2** — When finding the mean difference, μ_D, between values in a matched pair, it is important to define the order of subtraction.
- **LO VAR-7.D** — Verify the conditions for the test for a population mean, including the mean difference between values in matched pairs. [Skill 4.C]
- **EK VAR-7.D.1** — In order to make statistical inferences when testing a population mean, we must check for independence and that the sampling distribution is approximately normal: a. To check for independence: (i) Data should be collected using a random sample or a randomized experiment. (ii) When sampling without replacement, check that n $\leq$ 10% N.
- b. To check that the sampling distribution of $\bar{x}$ is approximately normal (shape): (i) If the observed distribution is skewed, n should be greater than 30. (ii) If the sample size is less than 30, the distribution of the sample data should be free from strong skewness and outliers.

Essential question: How does the study structure determine the parameter, hypotheses, and conditions for a mean test?

DOK-3 skill: 4.C · **FRQ pattern:** `paired-test-setup-and-consequences`

DOK rationale: Part (c) coordinates repeated measures, subtraction order, tail direction, independent sample size, and shape evidence, then traces the uncertainty and research-question consequences of the rejected setup.

Self-paced bonus sheet. Students take it when they choose and turn it in when they finish. Everything they need is printed on the sheet. Score part (c) only, E / P / I, by hand — never AI-graded or auto-scored. (a) and (b) are the ladder up; use them to see where a thin (c) came from.

Questions to ask

- Which specific number or study detail supports your choice?
- What claim would go beyond the evidence, and why?

[WARN] LOOK FOR

- Choosing a speaker without a reason.
- Copying numbers without explaining what they support.

SCORING PART (c) — E / P / I

Expected elements

- **paired-units** (required) — Justifies a one-sample test of the 18 within-technician differences
- **correct-hypotheses** (required) — Uses the population mean difference, zero null, and lower-tail alternative
- **rejects-independent-times** (required) — Explains that the two times from each person are linked

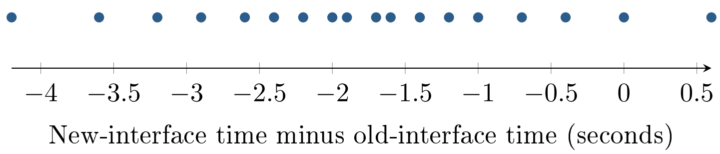
E	Justifies the paired units and explains the lower-tail population hypotheses using new minus old and the reduction question.
P	Shows substantive valid reasoning for at least one required element, but does not meet all E requirements. Credit correct reasoning even when another claim is wrong.
I	Shows no substantive valid reasoning for any required element; a name, unsupported choice, or copied number alone does not count.

[STAR] THE PROBLEM — annotated

Suppose a company takes an SRS of 18 of its 300 technicians. Each completes the same task with old and new interfaces; interface order is randomly assigned. The plot shows $d = \text{new time} - \text{old time}$. Researchers ask whether the new interface reduces the population mean time.

Nia: "The technician seems to be the independent unit, so I would build one difference per person; the subtraction order matters for the tail."

Colin: "There are 36 recorded times, so I would use two independent samples of 18 and a two-sided alternative."



FIRST TAKE — one sentence before you work the parts

Would you compare each technician's two times or treat them as separate people? Explain in one sentence.
Key: Ungraded commitment; accept any evidence-based first impression.

[BOOK] SET UP A TEST FOR A MEAN (keep on the page)

For a population mean with unknown population standard deviation, use a one-sample t-test. Matched pairs are two linked measurements: subtract in the same order and analyze one sample of differences. Write hypotheses about the population mean, not the sample mean. The null uses equality; the alternative uses $<$, $>$, or $\neq$, as determined by the research question and subtraction order. Check random sampling or assignment, $n \leq 0.10N$ when sampling without replacement, and shape. For $n < 30$, the values being analyzed should have no strong skewness or outliers.

DOK 1 **(a)** Define μ_d in context.

Key: μ_d is the mean new-minus-old task time for all 300 technicians.

DOK 2 **(b)** Check the random-sample, 10 percent, and small-sample shape conditions in three short notes.

Key: The 18 technicians were randomly sampled. $18 \leq 0.10(300) = 30$. Since $n = 18 < 30$, inspect the differences: the plot shows no strong skewness or outliers.

DOK 3 **(c)** In 3–4 sentences, choose a setup and justify it. State the hypotheses using the given subtraction order and reduction question, then explain why the 36 times are not 36 independent people.

Key: Nia is justified: use a matched-pairs one-sample t-test on the 18 differences. For new minus old, a reduction is negative, so $H_0 : \mu_d = 0$ and $H_a : \mu_d < 0$. Each technician supplies two linked times, so there are 18 independent people, not 36. Treating the times as independent loses the pairing, and a two-sided alternative asks about any change rather than the planned reduction.